

# Measuring Incremental ROI of Sponsored Search Advertising: A Difference-in-Differences Analysis

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## Executive Summary

Bazaar.com aims to evaluate the true return on investment (ROI) of its sponsored search advertising on branded keywords. While initial calculations based on paid clicks suggest a high ROI, this approach does not distinguish between total traffic and incremental traffic, potentially overstating the value of advertising.

To address this issue, we leverage a natural experiment created by the temporary suspension of Google sponsored ads in weeks 10–12. Using a difference-in-differences framework with other search platforms as a control group, we estimate the causal impact of sponsored ads on total traffic.

The results show that the removal of sponsored ads leads to a substantial decline in Google traffic, indicating that advertising does generate incremental visits. However, a simple pre-post comparison understates this effect because it fails to account for broader time trends. Difference-in-differences estimates indicate that suspending Google sponsored ads reduced weekly traffic by approximately 9,911 visits relative to the control platforms.

Using this estimate, we revise Bob’s ROI calculation by focusing on incremental traffic rather than total paid clicks. The revised ROI provides a more causally grounded measure of advertising effectiveness because it is based on estimated incremental visits rather than observed paid clicks. However, the exact magnitude should be interpreted with caution given the scaled nature of the dataset.

Given the positive incremental impact, Bazaar should maintain investment in sponsored search advertising, while adopting a more targeted strategy—reducing inefficient branded keyword spending, prioritizing segments where ads are more likely to generate incremental traffic, and evaluating performance based on incremental impact rather than raw click-based metrics.

## Business Context

Bazaar.com is a large online retailer that uses paid search advertising on platforms such as Google and Bing. The company bids on branded keywords that include its own name, such as searches containing “Bazaar,” which typically indicate strong user intent to visit the website.

When users search for branded terms, they are shown both sponsored ads and organic search results linking to Bazaar.com. Because these users are already aware of the brand and actively searching for it, they are likely to visit the website even without the presence of paid ads.

This creates a key business challenge: it is unclear whether sponsored ads for branded keywords generate additional traffic or simply capture users who would have arrived through organic search anyway. Bob’s analysis assumes that all paid clicks are incremental, while Myra questions this assumption and suggests that many of these visits would have occurred regardless of advertising.

This distinction is critical because it directly affects how the company measures the return on its advertising spend. If a large portion of paid traffic is not incremental, then the true value of sponsored search advertising may be significantly lower than initially estimated.

## Why Click-Based ROI Is Biased

Bob's calculation is problematic because it confuses **observed paid clicks** with **incremental clicks caused by advertising**.

His approach assumes that:

1. every sponsored click represents a new visitor generated by the ad, and
2. without the ad, these users would not have visited Bazaar.com.

These assumptions are unlikely to hold, especially for branded keyword searches. Many users searching for Bazaar.com already have strong intent to visit the website. If the sponsored ad were removed, a large share of these users would likely click on the organic link instead.

As a result, the ad often changes the channel of arrival (paid vs. organic) rather than generating truly new traffic. This substitution effect means that not all paid clicks are incremental.

At a deeper level, Bob's calculation fails to consider the counterfactual scenario, meaning what would have happened in the absence of advertising. Because it ignores this, the analysis overestimates the true causal impact of ads.

Therefore, Bob is measuring the profitability of paid clicks, rather than the profitability of incremental traffic, leading to an overestimation of ROI.

## Treatment and Control Design

```
ads <- ads %>%
  mutate(
    platform = tolower(platform),
    week = as.integer(week),
    after = ifelse(week >= 10, 1, 0),
    treated = ifelse(platform == "goog", 1, 0),
    total_traffic = avg_spons + avg_org,
  )
```

The unit of observation in this dataset is a **platform-week observation**.

The treatment is **the temporary cessation of Google sponsored search advertising beginning in week 10 due to a technical glitch**. **Google is the treated unit** because its sponsored ads were interrupted during this period. In contrast, **Bing, Yahoo, and Ask serve as control units** because their sponsored ads continued to run, providing a counterfactual for what would have happened to Google traffic in the absence of the interruption.

We define: - after = 1 for weeks 10–12 and 0 for weeks 1–9 - treated = 1 for Google and 0 for Bing, Yahoo, and Ask

This setup forms a classic difference-in-differences design, where one unit experiences a treatment shock while comparable units do not. The treatment effect is captured by the interaction between the treated unit and the post-treatment period.

This design allows us to estimate the causal effect by comparing the change in Google traffic before and after the interruption to the corresponding change in the control platforms, which serve as a counterfactual.

## Overall Traffic Trends

```
traffic_plot <- ads %>%
  mutate(
    platform = case_when(
      platform == "goog" ~ "Google",
```

```

platform == "bing" ~ "Bing",
platform == "yahoo" ~ "Yahoo",
platform == "ask" ~ "Ask",
TRUE ~ platform
)
)

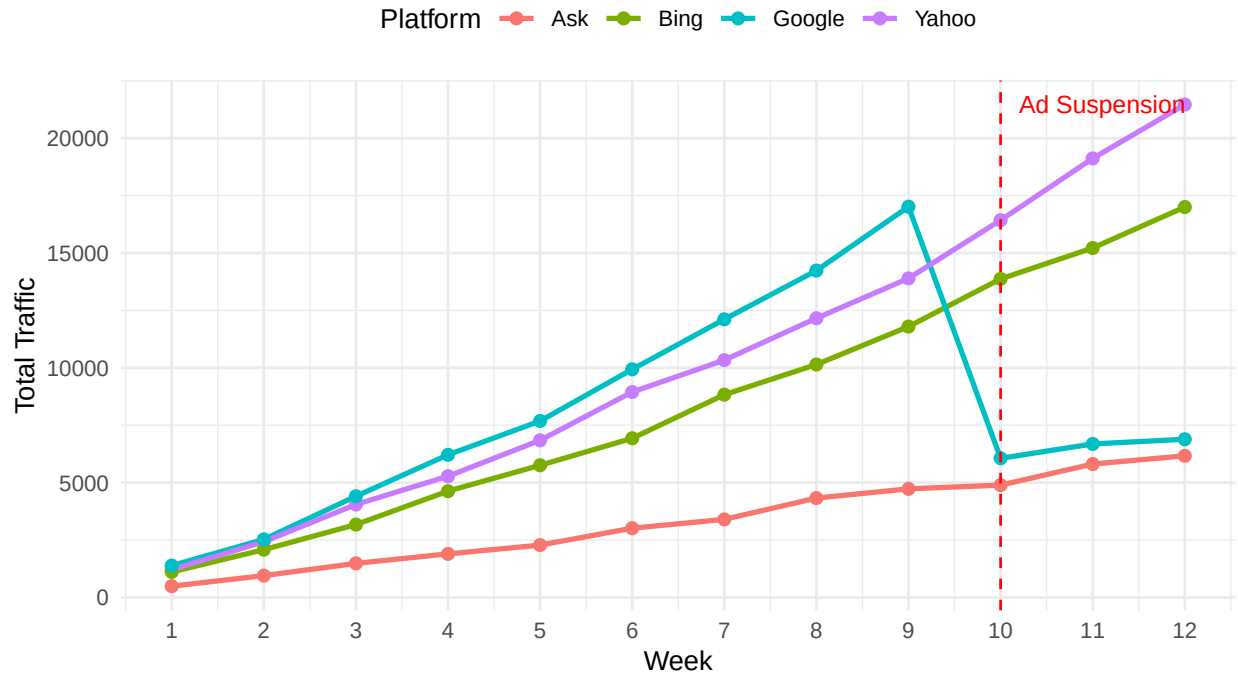
overall_plot <- ggplot(
  traffic_plot,
  aes(
    x = week,
    y = total_traffic,
    color = platform,
    group = platform
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_vline(
    xintercept = 10,
    linetype = "dashed",
    color = "red"
  ) +
  annotate(
    "text",
    x = 10.2,
    y = max(traffic_plot$total_traffic),
    label = "Ad Suspension",
    color = "red",
    hjust = 0,
    size = 3.5
  ) +
  scale_x_continuous(breaks = 1:12) +
  labs(
    title = "Traffic Trends Across Search Platforms",
    subtitle = "Google sponsored ads were suspended during Weeks 10-12",
    x = "Week",
    y = "Total Traffic",
    color = "Platform"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top"
  )
)

overall_plot

```

## Traffic Trends Across Search Platforms

Google sponsored ads were suspended during Weeks 10–12



```
ggsave(  
  "figures/overall_traffic_trends.png",  
  plot = overall_plot,  
  width = 8,  
  height = 5,  
  dpi = 300,  
  create.dir = TRUE  
)
```

Before estimating the causal effect, we first examine traffic patterns across all search platforms over time. Weeks 10–12 correspond to the period during which Google sponsored ads were temporarily suspended because of a technical glitch.

The figure shows that traffic across all platforms generally follows an upward trend over time. However, Google exhibits a noticeable decline immediately after the advertising interruption, whereas Bing, Yahoo, and Ask continue their previous trajectories. This visual evidence provides an initial indication that the reduction in Google traffic is associated with the suspension of sponsored ads rather than an overall market slowdown. This motivates the use of a difference-in-differences design to estimate the causal effect while controlling for common time trends.

## Parallel Trends Check

```
pre_trend <- ads %>%  
  filter(week <= 9) %>%  
  mutate(  
    platform = case_when(  
      platform == "goog" ~ "Google",  
      platform == "bing" ~ "Bing",
```

```

    platform == "yahoo" ~ "Yahoo",
    platform == "ask" ~ "Ask",
    TRUE ~ platform
  )
)

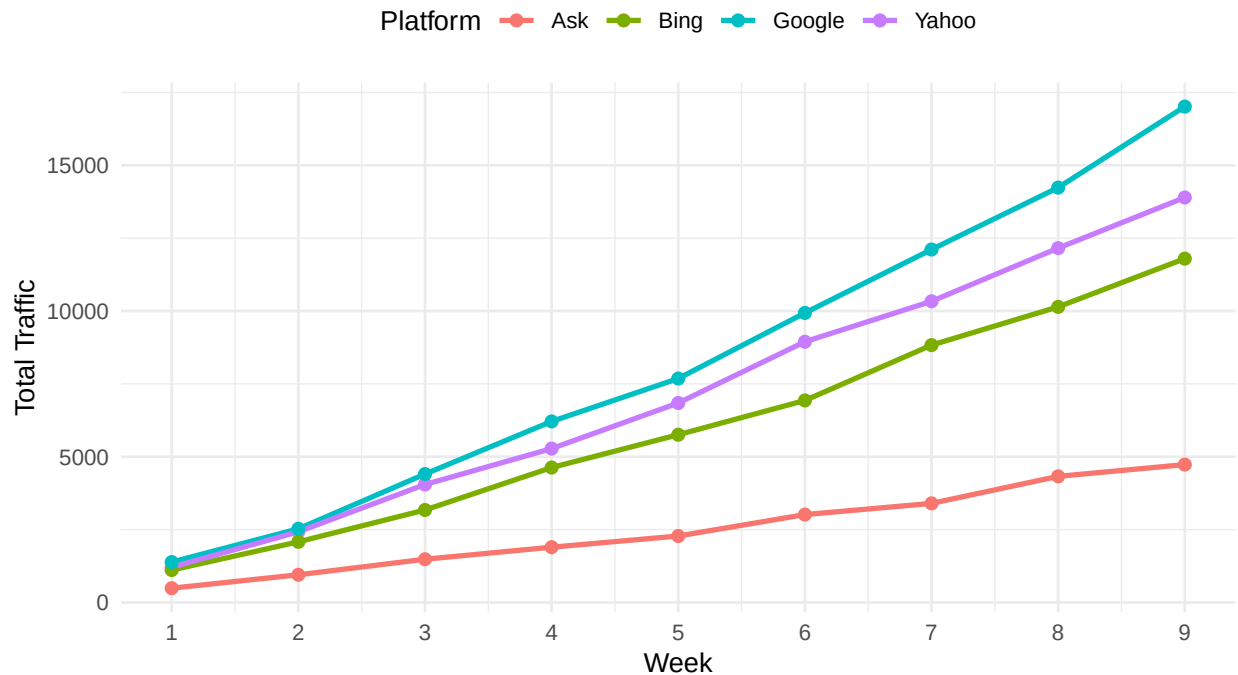
parallel_plot <- ggplot(
  pre_trend,
  aes(
    x = week,
    y = total_traffic,
    color = platform,
    group = platform
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 1:9) +
  labs(
    title = "Pre-Treatment Traffic Trends by Search Platform",
    subtitle = "Traffic trends before the Google sponsored ad suspension (Weeks 1-9)",
    x = "Week",
    y = "Total Traffic",
    color = "Platform"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top"
  )

parallel_plot

```

## Pre-Treatment Traffic Trends by Search Platform

Traffic trends before the Google sponsored ad suspension (Weeks 1–9)



```
ggsave(  
  "figures/parallel_trends.png",  
  plot = parallel_plot,  
  width = 8,  
  height = 5,  
  dpi = 300,  
  create.dir = TRUE  
)
```

Before applying the difference-in-differences model, we visually compare traffic trends across Google, Bing, Yahoo, and Ask during the pre-treatment period (weeks 1–9). Although the platforms differ in traffic volume, all exhibit steadily increasing trends before the advertising suspension. The comparison focuses on the similarity of trends rather than traffic levels, since the parallel trends assumption concerns changes over time rather than absolute values. No obvious structural break or abrupt divergence is observed prior to treatment, providing some support for the parallel trends assumption. While this visual inspection cannot formally verify the assumption, it suggests that the control platforms provide a reasonable counterfactual for Google.

## First-Difference Estimate

```
# Keep only the treated unit: Google  
google_only <- ads %>%  
  filter(platform == "goog")  
  
# Simple pre-post regression  
post_model <- lm(total_traffic ~ after, data = google_only)  
summary(post_model)
```

```
##
## Call:
## lm(formula = total_traffic ~ after, data = google_only)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7003.9 -2630.1  -172.5   2088.4  8625.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8390         1598   5.252 0.000373 ***
## after          -1846         3195  -0.578 0.576238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4793 on 10 degrees of freedom
## Multiple R-squared:  0.0323, Adjusted R-squared:  -0.06447
## F-statistic: 0.3337 on 1 and 10 DF,  p-value: 0.5762

# Extract coefficient from the regression
post_coef <- coef(post_model)[["after"]]

# Pre and post means
pre_google <- mean(google_only$total_traffic[google_only$after == 0])
post_google <- mean(google_only$total_traffic[google_only$after == 1])

# First-difference estimates
first_diff_level <- post_google - pre_google
first_diff_pct <- (post_google - pre_google) / pre_google

first_diff_level

## [1] -1845.889

first_diff_pct

## [1] -0.2200135
```

## Results and Interpretation

The simple pre-post regression estimates the treatment effect as the coefficient on after, which is -1846. This indicates that average weekly Google traffic decreased by approximately 1,846 visits after week 10.

In percentage terms, average traffic declined from 8,390 to 6,544.00, corresponding to a 22.0% decrease.

## Why This Estimate Is Not Causal

Although this estimate suggests a substantial decline in traffic, it should not be interpreted as the causal effect of the ad interruption. The pre-post comparison assumes that the only change between the two periods is the removal of sponsored ads.

This assumption is unlikely to hold, as traffic may also be affected by time trends, seasonality, or other market-wide factors. As a result, the estimate may confound the true treatment effect with these external changes.

Therefore, the first-difference estimate is only a descriptive measure and does not provide a reliable causal estimate. To isolate the true effect of the ads, we need to compare this change to a control group, which motivates the use of a difference-in-differences approach.

## Difference-in-Differences Estimate

```
# Difference-in-Differences regression
did_model <- lm(total_traffic ~ treated + after + treated:after, data = ads)
summary(did_model)

##
## Call:
## lm(formula = total_traffic ~ treated + after + treated:after,
##     data = ads)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8437.7 -3231.0 -510.5  3591.6  8630.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5265.0      882.5   5.966 3.79e-07 ***
## treated        3124.9     1765.0   1.770  0.08357 .
## after          8064.7     1765.0   4.569 3.94e-05 ***
## treated:after -9910.6     3530.0  -2.808  0.00741 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4586 on 44 degrees of freedom
## Multiple R-squared:  0.3274, Adjusted R-squared:  0.2816
## F-statistic: 7.141 on 3 and 44 DF,  p-value: 0.0005211

# Extract treatment effect (DiD estimate)
did_coef <- coef(did_model)[["treated:after"]]

did_coef

## [1] -9910.593
```

**Natural Experiment and Identification Assumptions** The temporary suspension of Google sponsored ads due to a technical glitch provides a natural experiment. Because the interruption was not driven by a strategic decision, it can be considered plausibly exogenous.

To interpret the difference-in-differences estimate as causal, several assumptions must hold. First, the parallel trends assumption requires that, in the absence of the treatment, Google traffic would have evolved in parallel with the control platforms. This implies that the change observed in the control group captures the underlying time trend. Second, the no interference assumption (SUTVA) requires that the treatment applied to Google does not affect traffic on the control platforms.

Under these assumptions, the control platforms provide a valid counterfactual for what would have happened to Google traffic without the treatment.

**Results and Interpretation** We estimate a difference-in-differences regression of total traffic on treatment status, time period, and their interaction. The coefficient on the interaction term captures the causal effect of the suspension of sponsored ads.

The estimated treatment effect is -9,910.59, indicating that Google traffic decreased by approximately 9,911 visits per week relative to the change observed in the control platforms after the ads were suspended.

**Comparison with the Pre-Post Estimate** The first-difference estimate suggested a decline of 1,846 visits per week, or about 22%. However, the difference-in-differences estimate is substantially larger in magnitude.

This difference indicates that the pre-post estimator fails to account for time trends that also affect the control platforms. As a result, it does not accurately capture the true effect of the treatment.

### Implications for the Post Estimator

The pre-post approach attributes all observed changes in Google traffic to the removal of sponsored ads, ignoring concurrent changes in the broader environment. In contrast, the difference-in-differences approach uses the control group to isolate the common time trend, providing a more credible estimate of the causal effect.

### Difference-in-Differences Visualization

```
did_plot <- ads %>%
  mutate(
    total_traffic = avg_spons + avg_org,
    group = ifelse(platform == "goog", "Treated (Google)", "Control")
  ) %>%
  group_by(platform, group, after) %>%
  summarise(avg_platform = mean(total_traffic), .groups = "drop") %>%
  group_by(group, after) %>%
  summarise(avg_traffic = mean(avg_platform), .groups = "drop") %>%
  mutate(
    period = ifelse(after == 0, "Before", "After"),
    period = factor(period, levels = c("Before", "After"))
  )

did_plot_fig <- ggplot(
  did_plot,
  aes(x = group, y = avg_traffic, fill = period)
) +
  geom_bar(
    stat = "identity",
    position = position_dodge(width = 0.7),
    width = 0.6
  ) +
  geom_text(
    aes(label = round(avg_traffic, 0)),
    position = position_dodge(width = 0.7),
    vjust = -0.3,
    size = 4
  ) +
  scale_fill_manual(
    values = c(
      "Before" = "#00BFC4",
      "After" = "#F8766D"
    )
  ) +
  expand_limits(
    y = max(did_plot$avg_traffic) * 1.15
  ) +
  labs(
    title = "Difference-in-Differences Visualization",
    subtitle = "Comparison of treated and control groups before and after treatment",
    x = "",
    y = "Average Traffic",
    fill = ""
  )
```

```

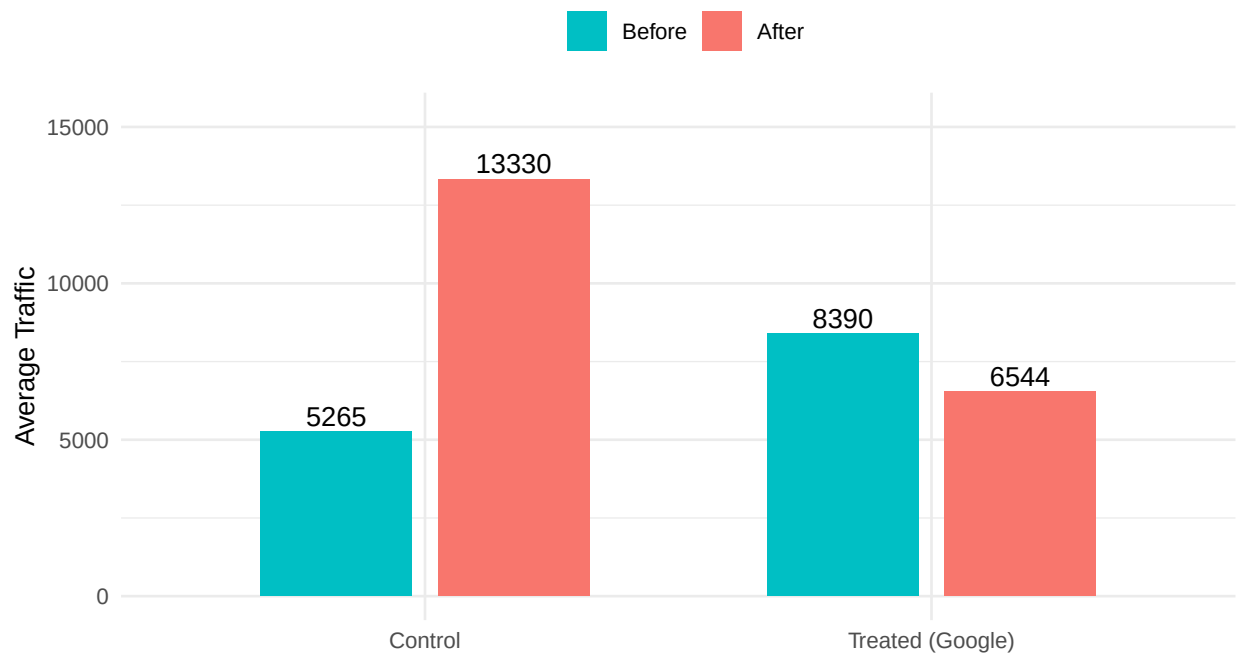
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  legend.position = "top"
)

did_plot_fig

```

## Difference-in-Differences Visualization

Comparison of treated and control groups before and after treatment



```

ggsave(
  "figures/did_visualization.png",
  plot = did_plot_fig,
  width = 7,
  height = 5,
  dpi = 300,
  create.dir = TRUE
)

```

The figure compares the average traffic of the treated group (Google) and the control group before and after the suspension of sponsored ads. The control group shows a substantial increase in traffic over time, rising from 5,265 to 13,330, which reflects the underlying upward time trend in search activity.

In contrast, the treated group experiences a decline in traffic from 8,390 to 6,544 after the ads are suspended. This suggests that, while overall traffic in the market is increasing, Google traffic moves in the opposite direction during the same period.

The difference between these two changes represents the difference-in-differences estimate. Specifically, the treated group decreases while the control group increases, implying a large negative treatment effect. This visual evidence supports the regression result that the suspension of sponsored ads led to a substantial

reduction in Google traffic relative to the counterfactual trend.

## Understanding the Treatment Effect

```
# DiD for sponsored traffic
did_spons <- lm(avg_spons ~ treated + after + treated:after,
               data = ads)

# DiD for organic traffic
did_org <- lm(avg_org ~ treated + after + treated:after,
              data = ads)

# Extract treatment effects
spons_effect <- coef(did_spons)[["treated:after"]]
org_effect   <- coef(did_org)[["treated:after"]]
total_effect <- coef(did_model)[["treated:after"]]

# Organic recovery rate
organic_recovery_rate <- abs(org_effect) / abs(spons_effect)

# Summary table
decomposition_tbl <- tibble(
  Metric = c(
    "Sponsored Traffic Effect",
    "Organic Traffic Effect",
    "Net Incremental Traffic",
    "Organic Recovery Rate"
  ),
  Estimate = c(
    comma(round(spons_effect)),
    comma(round(org_effect)),
    comma(round(total_effect)),
    percent(organic_recovery_rate, accuracy = 1)
  )
)

kable(
  decomposition_tbl,
  col.names = c("Metric", "Estimate")
)
```

| Metric                   | Estimate |
|--------------------------|----------|
| Sponsored Traffic Effect | -12,204  |
| Organic Traffic Effect   | 2,293    |
| Net Incremental Traffic  | -9,911   |
| Organic Recovery Rate    | 19%      |

```
# Decomposition visualization
decomp_plot <- tibble(
  Traffic = factor(
    c(
      "Sponsored\nTraffic",

```

```

      "Organic\nTraffic",
      "Net\nIncremental\nTraffic"
    ),
    levels = c(
      "Sponsored\nTraffic",
      "Organic\nTraffic",
      "Net\nIncremental\nTraffic"
    )
  ),
  Effect = c(
    spons_effect,
    org_effect,
    total_effect
  )
)

decomp_plot_fig <- ggplot(
  decomp_plot,
  aes(
    x = Traffic,
    y = Effect,
    fill = Traffic
  )
) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    color = "gray60",
    linewidth = 0.5
  ) +
  geom_col(width = 0.6) +
  geom_text(
    aes(label = comma(round(Effect))),
    vjust = ifelse(decomp_plot$Effect > 0, -0.6, 1.3),
    size = 4.2,
    fontface = "bold"
  ) +
  annotate(
    "text",
    x = 2.55,
    y = 1800,
    label = "~19% organic recovery"

    ,
    size = 3.6,
    color = "gray30",
    hjust = 0
  ) +
  scale_fill_manual(
    values = c(
      "Sponsored\nTraffic" = "#D55E00",
      "Organic\nTraffic" = "#009E73",
      "Net\nIncremental\nTraffic" = "#0072B2"
    )
  )

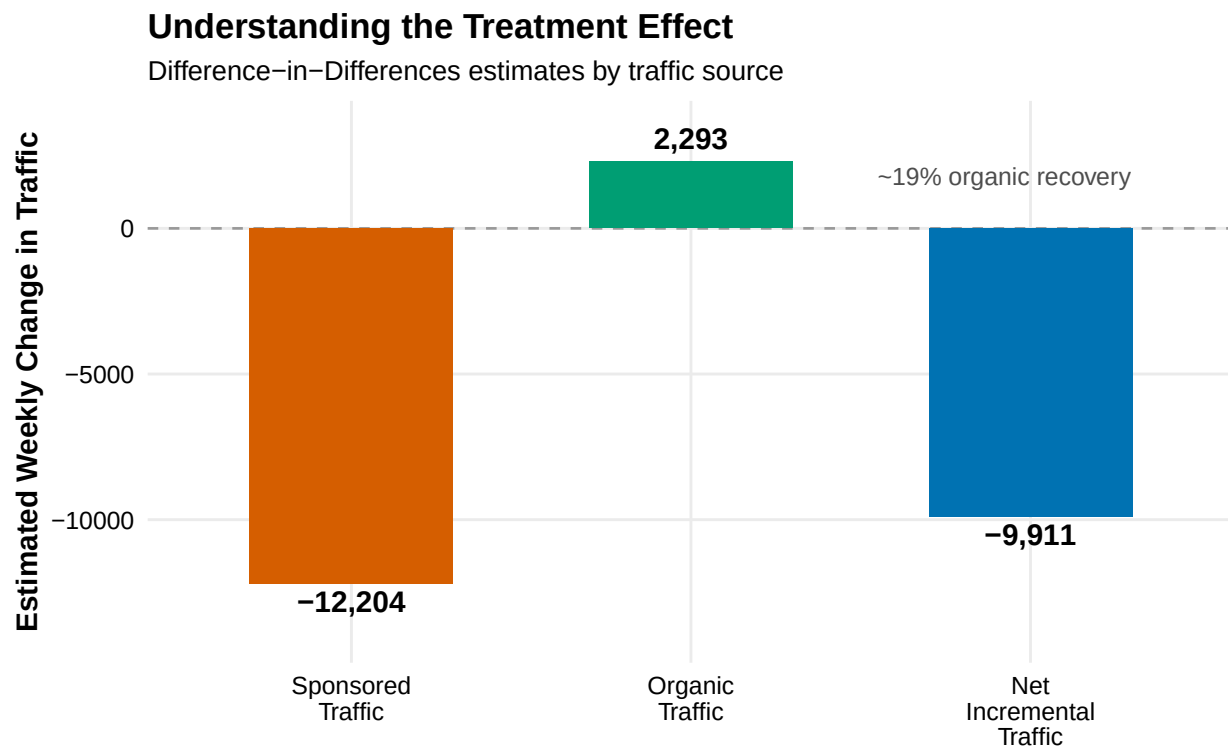
```

```

)
) +
coord_cartesian(ylim = c(min(decomp_plot$Effect) * 1.15, 3500)) +
labs(
  title = "Understanding the Treatment Effect",
  subtitle = "Difference-in-Differences estimates by traffic source",
  x = "",
  y = "Estimated Weekly Change in Traffic"
) +
theme_minimal(base_size = 12) +
theme(
  legend.position = "none",
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(size = 11),
  axis.title.y = element_text(face = "bold"),
  axis.text = element_text(color = "black"),
  panel.grid.minor = element_blank()
)

```

decomp\_plot\_fig



```

ggsave(
  "figures/treatment_effect_decomposition.png",
  plot = decomp_plot_fig,
  width = 8,
  height = 5,
  dpi = 300,
  create.dir = TRUE
)

```

)

The overall treatment effect can be decomposed into two offsetting components. When Google sponsored ads were suspended, sponsored traffic decreased by approximately 12,204 visits per week, while organic traffic increased by about 2,293 visits per week. This indicates that some users substituted from paid advertisements to organic search results instead of leaving the website entirely.

The estimated organic recovery rate is approximately 19%, meaning that about one-fifth of the lost sponsored traffic was recovered through organic search. However, this recovery was not sufficient to offset the decline in sponsored traffic, resulting in a net loss of approximately 9,911 weekly visits. These findings suggest that sponsored ads generate meaningful incremental traffic rather than simply shifting users between paid and organic channels, reinforcing the importance of evaluating advertising performance using incremental impact instead of raw paid clicks.

## ROI Correction Using Incremental Traffic

Bob's ROI is overstated because it assumes all paid clicks are incremental. In reality, many users would have visited through organic search even without ads. Therefore, we use the DiD estimate to measure incremental traffic. From the DiD results, sponsored ads generate approximately 9,911 incremental visits per week.

```
incremental_visits <- abs(did_coef)
conversion_rate <- 0.12
margin_per_conversion <- 21
cost_per_click <- 0.60

avg_paid_clicks <- mean(google_only$avg_spons[google_only$after == 0])

incremental_conversions <- incremental_visits * conversion_rate
incremental_margin <- incremental_conversions * margin_per_conversion
advertising_cost <- avg_paid_clicks * cost_per_click
incremental_roi <- (incremental_margin - advertising_cost) / advertising_cost
```

Using the DiD estimate, the corrected ROI is calculated as follows:

- Incremental visits: 9,911
- Conversion rate: 12%
- Incremental conversions: 1,189
- Contribution margin: \$24,975
- Advertising cost: \$3,674
- Estimated incremental ROI: **580%**

## Managerial Recommendations

The analysis shows that sponsored search advertising does generate incremental traffic, as total visits decline when ads are suspended. This indicates that paid ads play a role in driving additional customer activity rather than simply shifting all traffic from organic channels.

At the same time, the exact ROI magnitude should be interpreted with caution because the analysis is based on a scaled dataset. Therefore, the key takeaway is not the precise ROI value, but the direction and business implication of the effect: sponsored ads appear to generate positive incremental traffic, and future marketing decisions should be based on incremental impact rather than raw click volume.

Given these findings, Bazaar should continue investing in sponsored search advertising, but adopt a more disciplined and data-driven approach. In particular, the company should monitor incremental performance rather than relying on raw click-based metrics, and periodically validate results through controlled experiments where possible.

In addition, Bazaar may consider selectively optimizing its ad spend, focusing on segments or keywords where the incremental impact is more likely to be meaningful, rather than applying a uniform strategy across all branded searches.

Overall, the evidence suggests that sponsored ads contribute positively to traffic and revenue, but careful measurement and optimization are necessary to ensure that marketing investments are aligned with true business value.